Yuan (Merlin) Chang

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Experience

- **AWS Annapurna Labs - Neuron Automated Reasoning Group** — Seattle, WA
  Software Development Engineer II (Compiler Engineer) — *Oct 2026 – Present*
  - Building a formally verified language stack for writing highly optimized vector engine kernels.
  - Building a formally verified assembler for the Trainium/Inferentia instruction set architecture (ISA).
  - Building simulation of AWS Trainium machine learning accelerator hardware.
- **Amazon Web Services (AWS) - Automated Reasoning Group** — Seattle, WA
  Software Development Engineer II — *Jul 2026 – Oct 2026*
  Software Development Engineer I — *Feb 2025 – Jul 2026*
  - Built an AI agent tool call evaluation system using formal policies such as Dogwood to improve agent safety.
  - Raised the compute fleet scaling ceiling by 20x (3k to 60k) and cut per-unit scale-up time by 60x through a backend migration of a highly concurrent, distributed service to Firecracker microVMs.
  - Lowered the barrier for internal builders and scientists to adopt formal reasoning engines (SAT/SMT, Lean4) by designing, implementing and shipping remote stateful MCP servers.
  - Eliminated outages from partial availability-zone dependency failures with Elastic Load Balancing over deep health checks, and the top recurring source of on-call pages with idempotent, order-tolerant ECS event handling.
  - Raised the team's operational bar with a fault injection framework, regular game days, and ordered, automated deployment sequencing that removed cross-pipeline dependencies.
  - Mentored junior engineers and interns through design and code review, scoping and guiding an intern project that converted to a full-time offer.
- **Amazon Web Services (AWS) - High Performance Computing** — Seattle, WA
  Software Development Engineer I — *Apr 2024 – Feb 2025*
  - Shipped AWS Parallel Computing Service (PCS), a public managed HPC service, as part of the launch team, building across the AWS Console frontend, the backend service APIs, and the workflows that provisioned customer compute clusters.

  Software Development Engineer Intern — *Jun 2023 – Sep 2023*
  - Designed internal RESTful APIs and orchestration that collapsed multi-step cluster configuration and job scheduling into single API invocations.
  - Shipped a public AWS CLI command exposing the managed HPC service directly to enterprise customers.
- **UC Davis Applied Mathematics Research Group** — Davis, CA
  Research Assistant & Software Engineer — *Jun 2021 – Jun 2022*
  - Developed automated reasoning pipeline that computed undiscovered Ramsey constants.
  - Reduced the search space by 6x through symmetry breaking and optimization techniques in preprocessing.
  - Achieved up to 10x improvement in SAT clause generation through computation optimizations.

Technical Skills

- **Languages** — Java, TypeScript, Bash, Rust, Python
- **Distributed Systems** — REST APIs, NoSQL, Firecracker microVMs, containerization (Docker), image-based deployment
- **Cloud & Build** — AWS (Lambda, ECS, SQS, DynamoDB, ElastiCache, S3, ELB), build systems, CI/CD tooling, Git
- **Agentic AI** — MCP servers, agent skills, steering docs, Claude Code, Kiro

Education

- **University of California, San Diego** — La Jolla, CA
  Master of Science - Computer Science and Engineering — *2022 – 2024*
- **University of California, Davis** — Davis, CA
  Bachelor of Science - Computer Science, Minor in Mathematics; Graduated with Honors — *2018 – 2022*

Publications

- **Rado Numbers and SAT Computations (with J. A. De Loera and W. J. Wesley).**
  Proceedings of the 47th International Symposium on Symbolic and Algebraic Computation (ISSAC 2022). Pages 333-342, available online at https://dl.acm.org/doi/10.1145/3476446.3535494.

Projects

- **Computations with Rado Numbers and Degree of Regularity (Automated Reasoning, Theory of Computation, Number Theory, Combinatorics)**: Advancement in terms of Rado Numbers and the degree of regularity. Discovery of nearly 500 new mathematical constants. Results collected and formulated into research paper accepted into 2022 ISSAC conference. Co-author with Prof. Jesús De Loera and William Wesley. Github Link
- **Minimization of CA Wildfire Impact (Convex Optimization, CVXPY)**: Prioritized fire suppression across 58 CA counties under budget constraints via the Lagrangian dual, KKT conditions, and a near-integral convex relaxation.
- **Terminal UI Board-game In Haskell (Haskell, Functional Programming, QuickCheck, Genetic Algorithm)**: TUI implementation of the classic board-game Mastermind in Haskell, using monadic and functional programming, with 10 QuickCheck property tests and a genetic-algorithm AI player built on minimax. Github Link